

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Project Title

Secure Document Vault: A Role-Based Secure Document Management System

Version Information

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Version History Table

Version	Date of Release	Pages Affected	Reason for Change	Signature
1.0	01/06/2026	All	First Release	
1.1	09/06/2026	All	Added diagrams, new functional requirements, non-functional requirements, dependencies, references and expanded requirements	
1.2	11/06/2026	Cover Page, TOC, Footer	Added Project Title on first page, Version History Table, Table of Content and Page Number	

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1. Introduction

1.1 Purpose

The Secure Document Vault is a web-based application designed to securely store, manage, and share digital documents within an organization. The system provides authentication, role-based access control, document upload/download functionality, audit logging, and secure document storage.

1.2 Scope

The system enables users to:

- Upload and manage documents securely.
- Access documents based on assigned roles.
- Maintain document confidentiality and integrity.
- Track user activities through audit logs.
- Prevent unauthorized access to sensitive documents.

1.3 Intended Audience

- System Administrators
 - Organization Employees
 - Project Evaluators
 - Developers and Maintenance Team
-

2. Overall Description

2.1 Product Perspective

Secure Document Vault is a centralized document repository developed as a web application using:

- Frontend: React.js
- Backend: Spring Boot
- Database: MySQL

2.2 Product Functions

- User Registration & Login
- JWT-Based Authentication
- Role-Based Access Control
- Document Upload
- Document Download
- Document Search
- Document Categorization
- Audit Logging

- User Management

2.3 User Classes

Administrator

- Manage users
- Assign roles
- View audit logs
- Manage documents

Employee/User

- Upload documents
- View permitted documents
- Download documents
- Search documents

2.4 Assumptions

- Users have internet access.
- Documents are stored on the server.
- MySQL database is available.

2.5 Operating Environment

Client Side

- Windows 10/11
- Google Chrome
- Microsoft Edge

Server Side

- Spring Boot Application Server
- Java 21
- MySQL 8.x

Network

- TCP/IP Communication
- Local Area Network(LAN)

2.6 Design and Implementation Constraints

- System shall be developed using Spring Boot.
- Frontend shall be implemented using React.js.
- MySQL shall be used as the database.
- JWT shall be used for authentication.
- REST APIs shall be used for communication.
- File size shall be restricted to 20 MB.
- Only authorized users can access protected documents.

2.7 Dependencies

- Java JDK 21
 - Maven
 - Node.js
 - MySQL Server
 - Spring Boot Framework
 - React Framework
 - Internet Connectivity
-

3. External Interface Requirements

3.1 User Interface

Login Page

- Username
- Password
- Login Button

Dashboard

- Upload Document
- View Documents
- Search Documents
- User Profile

Admin Panel

- Manage Users
- View Audit Logs
- Manage Access Rights

3.2 Hardware Interface

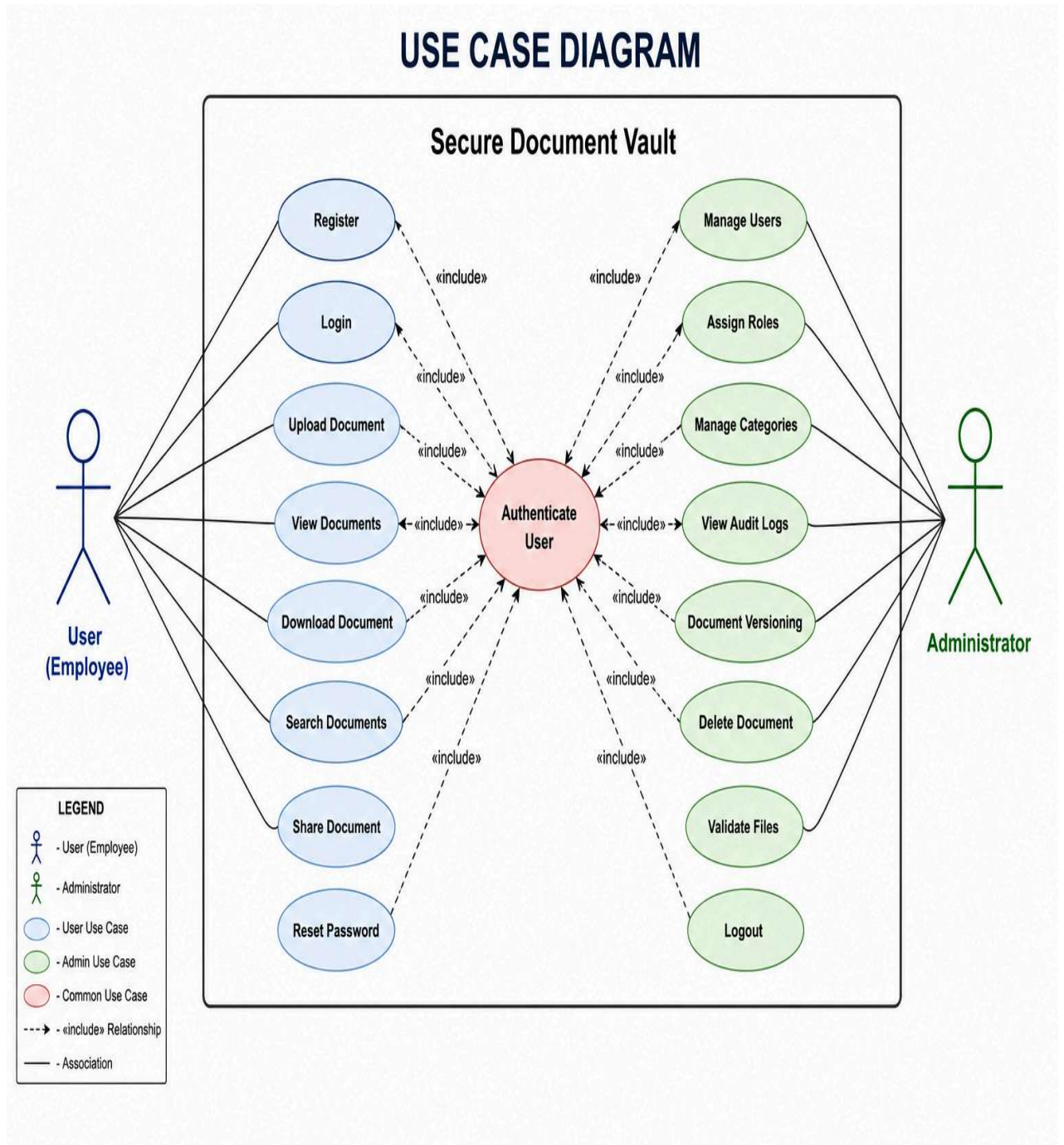
- Client PC/Laptop
- Server Machine
- Storage Device

3.3 Software Interface

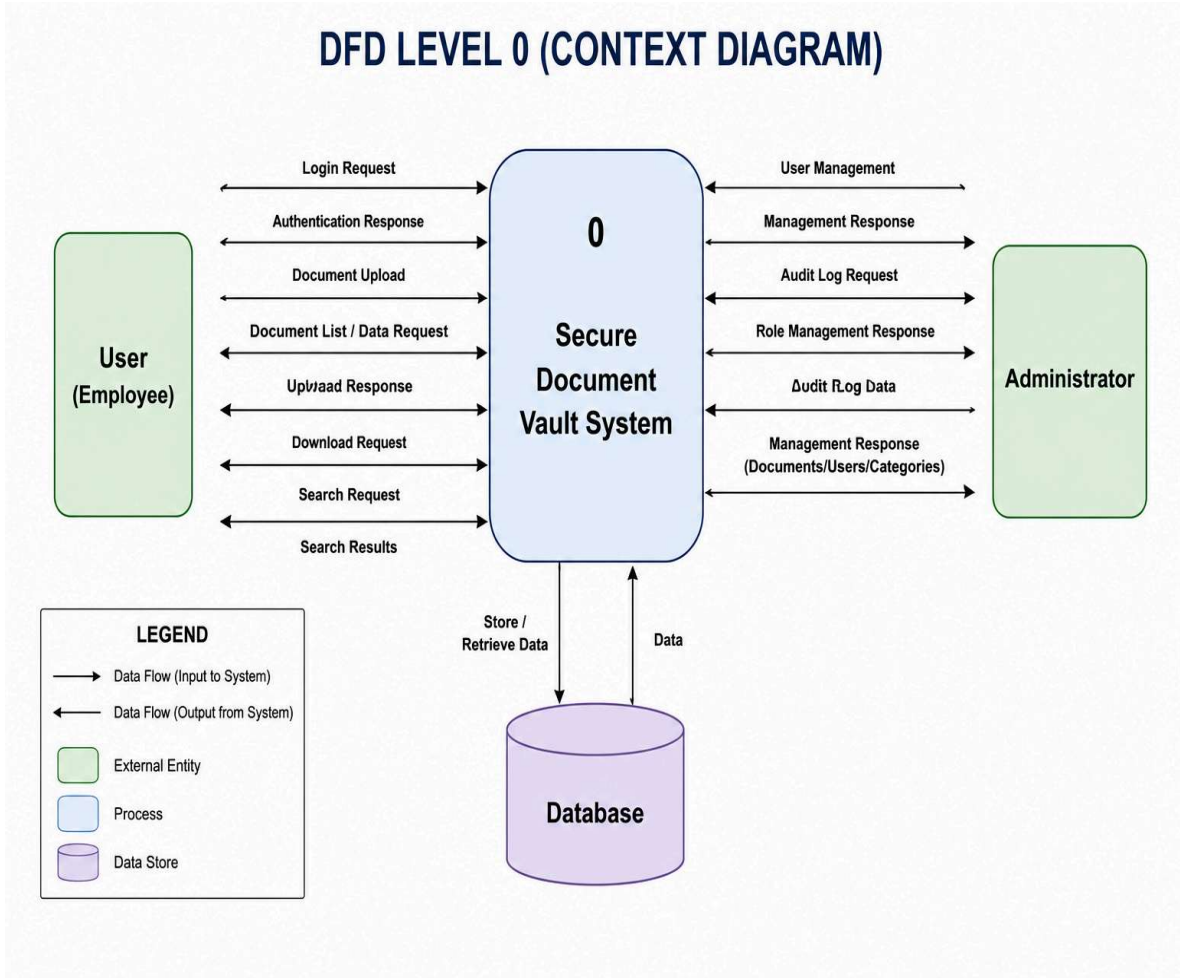
Component	Technology
Frontend	React.js
Backend	Spring Boot
Database	MySQL
Security	JWT
API	REST API

4. System Models and Diagrams

4.1 Use Case Diagram

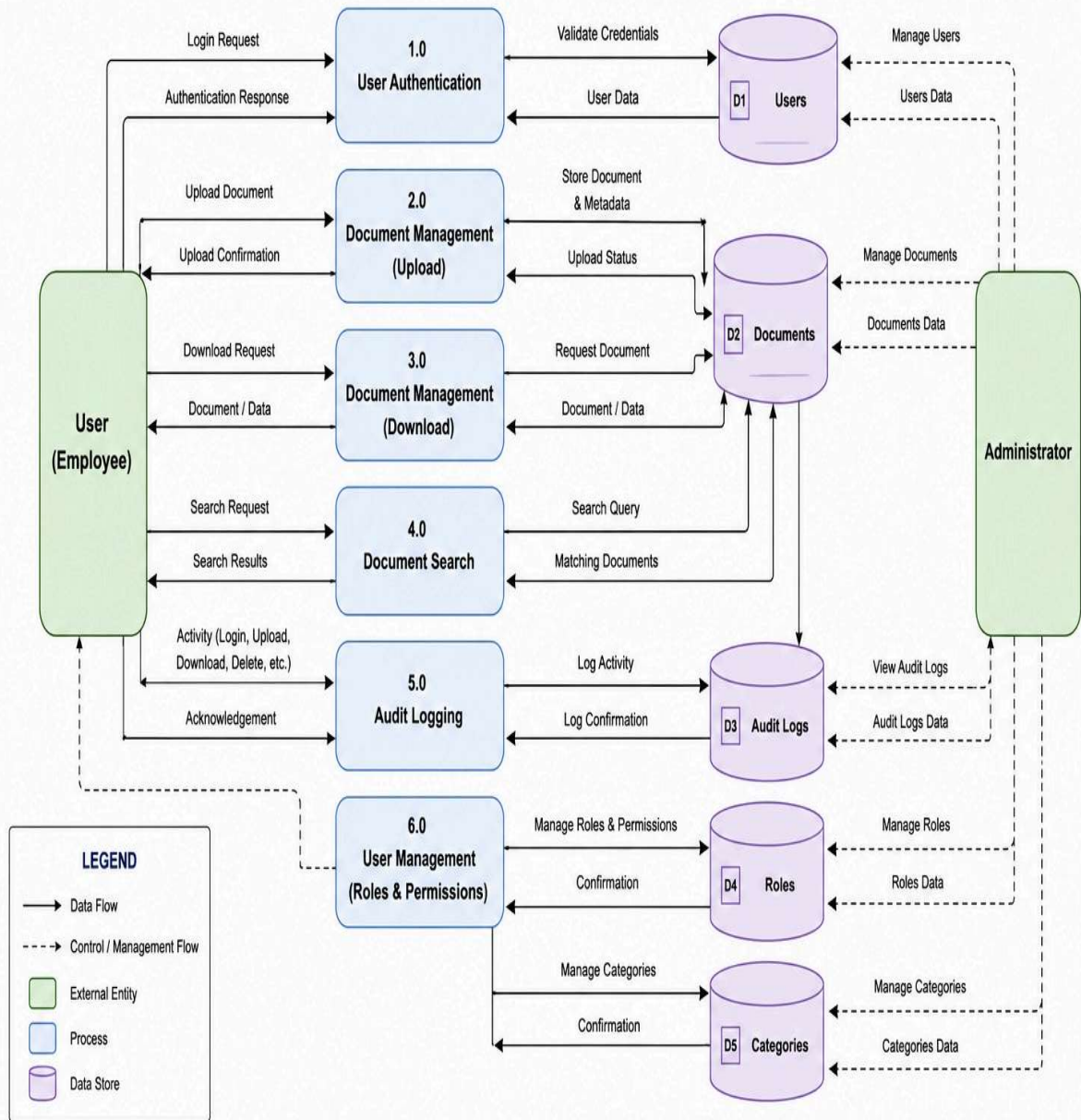


4.2 DFD Level 0

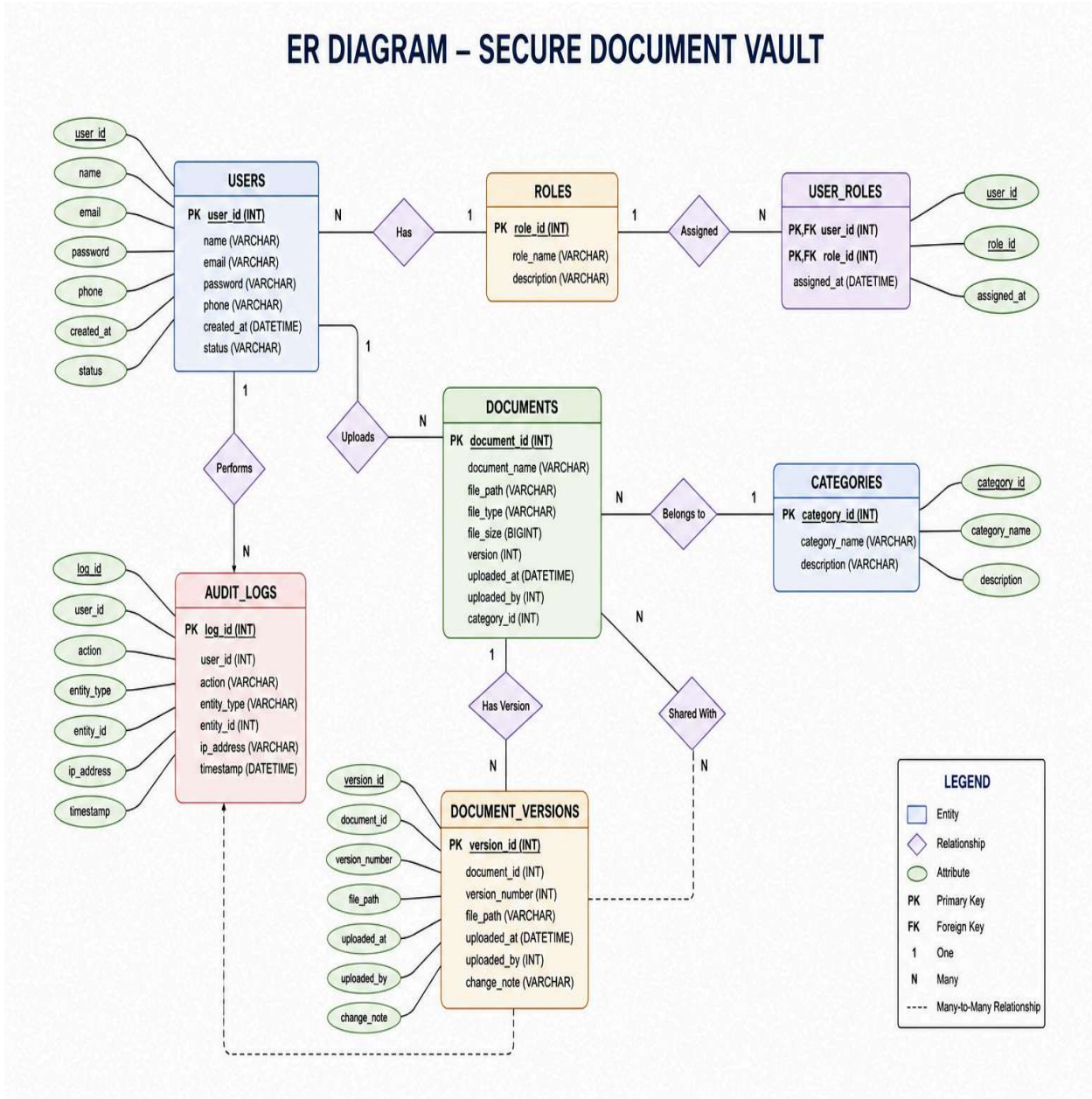


4.3 DFD Level 1

DFD LEVEL 1 DIAGRAM – SECURE DOCUMENT VAULT SYSTEM



4.4 ER Diagram



5. Functional Requirements

5.1 User Authentication

Description:

- Users authenticate using valid credentials.

Inputs:

- Email
- Password

Processing:

- Validate credentials
- Generate JWT token

Outputs:

- Login Success
- Login Failure
- JWT Token

5.2 Document Upload

Description

Allows authorized users to upload documents to the repository.

Inputs

- Document File
- Document Name
- Category

Processing

- Validate file type and size.
- Store files in storage.
- Save metadata in the database.
- Create audit log entry.

Outputs

- Upload Success Message
- Document Record Created

5.3 Document Download

Description

Allows authorized users to download documents.

Inputs

- Document ID

Processing

- Verify user permissions.
- Retrieve document from storage.
- Record download activity.

Outputs

- Downloaded Document
- Audit Log Entry

5.4 Role-Based Access Control

Description

Controls access based on assigned user roles.

Inputs

- User Role
- Requested Resource

Processing

- Verify permissions.
- Check role privileges.

Outputs

- Access Granted
- Access Denied

5.5 Document Search

Description

Allows users to search documents using keywords and filters.

Inputs

- Keyword
- Category
- Upload Date

Processing

- Search document metadata.
- Apply filters.

Outputs

- Matching Document List

5.6 Audit Logging

Description

Records all important system activities.

Inputs

- User Action
- System Event

Processing

- Capture activity details.
- Store log record.

Outputs

- Audit Log Entry

5.7 Document Versioning

Description

Maintains document history.

Inputs

- Existing Document
- Updated Document

Processing

- Create new version
- Store version information

Outputs

- Version Number
- Version History

5.8 Document Deletion

Description

Authorized users can delete documents.

Inputs

- Document ID

Processing

- Verify permissions
- Delete document

Outputs

- Deletion Confirmation
- Audit Entry

5.9 Forgot Password

Description

Allows password reset.

Inputs

- Registered Email

Processing

- Generate reset token
- Send email

Outputs

- Reset Link

5.10 File Validation

Description

Validates uploaded files.

Inputs

- Uploaded File

Processing

- Validate extension
- Validate size

Outputs

- Accepted
- Rejected

5.11 Document Sharing

Description

Shares documents with other users.

Inputs

- Document ID
- Recipient User

Processing

- Grant permission

Outputs

- Sharing Confirmation

6. Non-Functional Requirements

Performance

- Response time less than 10 seconds.

Security

- Password Encryption.
- JWT Authentication.
- Role-Based Authorization.

Reliability

- 99% availability.

Scalability

- Support future cloud deployment.

Maintainability

- Modular architecture.
- RESTful APIs.

Session Timeout

- System logs out inactive users after 15 minutes.

File Encryption at Rest

- Documents are encrypted using AES-256.

Audit Log Retention

- Audit logs retained for a minimum one year.
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7. References

- [1] Spring Boot Documentation - <https://spring.io/projects/spring-boot>
 - [2] React Documentation - <https://react.dev>
 - [3] MySQL Documentation - <https://dev.mysql.com/doc>
 - [4] OWASP Security Guidelines - <https://owasp.org>
 - [5] JWT RFC 7519 - <https://datatracker.ietf.org/doc/html/rfc7519>
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8. Acceptance Criteria

The project shall be accepted if:

- ✓ Users can register and login successfully.
 - ✓ Documents can be uploaded and stored securely.
 - ✓ Documents can be downloaded only by authorized users.
 - ✓ User activities are logged.
 - ✓ Admin can manage users and permissions.
 - ✓ Search functionality works correctly.
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9. Deliverables

1. Software Requirement Specification Document
2. Source Code
3. Database Schema
4. Test Cases
5. User Manual
6. Deployment Guide
7. Final Project Report
8. Presentation Slides